Aria Naraghi

Blockchain & Cryptography Researcher

aryaethn@gmail.com | Istanbul, Turkey | GitHub | LinkedIn | X / Twitter | Dune Analytics | PolyShield

PROFESSIONAL SUMMARY

Blockchain and cryptography researcher and builder with 7+ years of experience spanning integration analysis, DeFi protocol research, on-chain analytics, and applied cryptography. Currently leading integration research at Rango Exchange, a cross-chain DEX and bridge aggregator, having delivered 40+ in-depth reports directly informing co-founder decisions. Founder of PolyShield, a zero-knowledge privacy protocol for Polymarket live on Polygon mainnet. Author of three EIPs/ERCs and two independent cryptography research papers on arithmetization-oriented hashing and the quantum soundness of grinding in hash-based SNARKs. Active contributor to major open-source cryptography projects, including formally verified proof libraries in Lean4, Ethereum's post-quantum consensus client, and the Zcash and arkworks ecosystems. Deep expertise in zero-knowledge proofs (STARKs, Groth16, ZKML), lattice and post-quantum cryptography, FHE, and cross-chain intent architectures. Also an active technical writer and course instructor.

CORE COMPETENCIES

- Zero-Knowledge Proofs (ZKP)
- Blockchain Integration & Protocol Research
- Cross-Chain Architecture & Bridge Aggregation
- Smart Contract Development (Solidity)
- On-Chain Analytics (Dune / SQL)
- Cryptographic Primitive Design
- Formal Verification (**Lean 4**)
- Zero-Knowledge Circuit Development (Circom, Noir)
- Post-Quantum & Lattice-Based Cryptography
- DeFi Protocol Evaluation & Tokenomics
- Ethereum Improvement Proposals (EIP/ERC)
- Technical & Institutional-Grade Writing
- Python, **Rust**, Java, LaTeX
- Project Evaluation & Investment Research
- Pedagogy: Cryptography & Computer Science
- Copywriting & Content Creation (Blog, X/Twitter)
- Fully Homomorphic Encryption (FHE)

PROFESSIONAL EXPERIENCE

Founder & Protocol Engineer | **PolyShield** *2026 – Present | Remote*

- Founded and built PolyShield, a **zero-knowledge privacy vault** for Polymarket, end to end; the core protocol is live on Polygon mainnet with real deposits, bets, and withdrawals verified in production.
- Designed and implemented 9 **Circom/Groth16** zero-knowledge circuits (deposit, bet authorization, settlement credit, withdrawal, cancellation, position close, partial credit, consolidation), each verified on-chain by a dedicated **Solidity** verifier contract.
- Architected the full protocol stack: a UUPS-upgradeable **Solidity** vault with a Poseidon Merkle tree and nullifier registry, a **Node.js** signing and settlement layer, and a **Next.js**/Wagmi frontend that generates proofs client-side via WASM.
- Designed the system's privacy and threat model — wallet-derived secrets, nullifier-based double-spend protection, and wallet-bound withdrawal proofs — to eliminate on-chain linkability between deposits and bets.

Senior Blockchain Research Specialist | **Rango Exchange** *Jan 2025 – Present | Remote*

- Authored 30+ in-depth integration and protocol research reports covering layer-1/layer-2 chains, bridges, and DEX protocols, directly informing co-founder decisions at a 70+ chain aggregator.
- Designed and deployed the company's complete on-chain analytics presence — all dashboards and SQL queries tracking volume, routing performance, and cross-chain activity are published on the Rango Dune page.
- Authored the full catalogue of Rango Learn Center technical articles (as Arya .ETH), covering cryptographic concepts, DeFi mechanics, and cross-chain infrastructure.
- Designed and delivered an internal cryptography course for Rango colleagues, covering zero-knowledge proofs, cryptographic primitives, and their applications across DeFi and cross-chain contexts.
- Authored three Ethereum standards (EIP-7956, EIP-7998, ERC-7992).

Cryptocurrency Business Data Analyst | **Wallex** *Jun 2024 – Apr 2025 | Remote*

- Researched and analysed 50+ cryptocurrency projects, delivering actionable investment strategy insights to team leadership covering token positions, DeFi platforms, and fund hedging.
- Produced structured financial analysis reports evaluating risk-adjusted returns across governance tokens and liquidity pool positions.

Cryptocurrency Fundamental Researcher / R&D Team Lead | **Padash Sarmayeh** *Sep 2022 – Jun 2023 | Tehran, Iran*

- Analysed 20+ cryptocurrency projects; research directly contributed to a 30% AUM gain in live crypto markets.
- Promoted to R&D Team Lead to spearhead an Iranian-market blockchain product initiative, leading cross-functional team from ideation through early development.

Cryptocurrency Fundamental Researcher | **CoinPlus** *Dec 2021 – Oct 2022 | Tehran, Iran*

- Analysed 80+ cryptocurrency projects fundamentally and financially; research helped limit drawdown to ~20% during a 70% market-wide crash, significantly outperforming the benchmark.

Researcher | **Denxa***Sep 2021 – Sep 2022 | Remote*

- Evaluated 5+ blockchain networks for a cross-border money-exchange platform; conducted regulatory analysis across Canada, UAE, and 3+ additional jurisdictions. Cosmos SDK selected as final architecture.
- Researched tokenomics models across 100+ live cryptocurrency projects to design an optimal token economy for the platform.

Cryptocurrency Fundamental Researcher / R&D Team Lead | **Max Digital** *Feb 2021 – Sep 2021 | Tehran, Iran*

- Analysed 40+ cryptocurrency projects; research contributed to 50%+ AUM gains for the trading team.
- Promoted to R&D Team Lead for a blockchain supply-chain initiative, directing the team from concept through early-stage development.

Independent Trader & Researcher | **Personal** *Jan 2018 – Present*

- Managed a personal crypto portfolio (BTC-denominated), achieving 200%+ returns through systematic fundamental analysis and disciplined risk management.
- Conducted in-depth reviews of 500+ cryptocurrency projects across layer-1, layer-2, DeFi, decentralised storage, and decentralised insurance verticals.

TEACHING EXPERIENCE

Cryptography Instructor (Internal) | **Rango Exchange** *2025 | Remote*

- Designed and delivered a structured cryptography course for Rango colleagues covering ZK proofs, commitment schemes, BLS signatures, post-quantum cryptography, and practical applications in cross-chain and DeFi contexts.

Course Instructor | **FaraDars** *2022 – 2023 | Tehran, Iran*

- Recorded and published two offline courses on Iran's largest e-learning platform (4.5M+ learners) in Persian:
- Pine Script Programming Language for Creating Trader Bots in Financial Markets.
- Introduction to Cryptocurrency Fundamental Analysis (in partnership with Mazdax Digital Assets Exchange and CoinPlus) — taken by 500+ students.

Guest Lecturer – Introduction to Cryptocurrencies | **CoinPlus & Shahid Beheshti University** *2022 | Tehran, Iran*

- Delivered a live seminar on fundamental analysis and cryptocurrency project evaluation to 70+ students.

Teaching Assistant | **Sharif University of Technology** *Sep 2017 – Jun 2019 | Tehran, Iran*

- TA for five CS core courses: Basic/Advanced Programming, Data Structures, Computer Organization and Design, Algorithm Design and Analysis, and Languages Theory and Automata.

Informatics Olympiad Instructor | **Allameh Tabatabaei High School** *Nov 2016 – May 2018 | Tehran, Iran*

- Coached competitive programming students: 100% first-round pass rate, 50%+ second-round advancement, 20% third-round advancement; one student earned Gold, others Silver and Bronze medals.

PROJECTS, RESEARCH & ETHEREUM STANDARDS

EIP-7998 – Turn randao_reveal into a VRF — *Ethereum Improvement Proposal*

Transforms Ethereum's randao_reveal into a per-slot BLS-based Verifiable Random Function (VRF) by signing the previous epoch's RANDAO mix and current slot, providing unpredictable, verifiable per-slot randomness without changing validator duties.

EIP-7956 – Transaction Ordering via Block-level Randomness — *Ethereum Improvement Proposal*

Reduces MEV attack surface by enforcing deterministic transaction ordering derived from block-level randomness, removing validator discretion in intra-block transaction sequencing.

ERC-7992 – Verifiable ML Model Inference via ZKML — *Ethereum Request for Comments*

Defines minimal on-chain interfaces for registering ML model commitments and verifying zero-knowledge proofs of inference, enabling ERC-165 discoverable, trust-minimised AI inference on Ethereum.

ERC-PPAR – Privacy-Preserving Account Recovery — *Ethereum Request for Comments (Withdrawn)*

Proposed a privacy-preserving smart contract account recovery mechanism using ZK proofs to authenticate guardians without revealing their identities on-chain.

noir-rlwe-gadgets — *Verifiable Ring-LWE / BFV Encryption in Noir*

A **Noir** library of zero-knowledge gadgets proving that a BFV ciphertext is a correct, bounded-error encryption of a message, verified on-chain via UltraHonk — the first lattice-encryption gadget set built for the Noir ecosystem. Merged into the community's awesome-noir resource list.

htlp — *Homomorphic Time-Lock Puzzle*

A **Rust** implementation of Malavolta and Thyagarajan's linearly homomorphic time-lock puzzle (CRYPTO 2019), including a transparent, no-trusted-setup class-group variant with Wesolowski proofs of exponentiation. Demonstrated via a sealed-bid auction and a private-tally vote.

Chainfluence — *TON + TEE Trustless Advertising Marketplace*

A decentralised Telegram advertising marketplace built on TON blockchain with EigenCompute TEE-based execution. Features TonConnect ed25519 signature verification, TEE-managed deal-specific escrow via BIP-39/SLIP-0010 HD key derivation, and a minimal on-chain Deal Registry in Tolk. Stack: NestJS, Prisma, PostgreSQL, Telegram Bot API, GramJS, React/Vite, Docker Compose.

deBazaar — *Crypto Escrow Marketplace*

A decentralised escrow marketplace for digital and physical goods. Implements Pyth VRF-based random arbiter selection, structured arbitration flows, and verifiability segmentation by asset type. Supports trustless dispute resolution without centralised counterparty.

zkBacktest — *ZK-Verified Trading Strategy Backtester*

A backtesting framework that generates zero-knowledge proofs of strategy performance, allowing traders to verifiably demonstrate historical P&L results without revealing proprietary strategy logic. Designed for drawdown-aware, category-filtered systematic approaches.

OPEN-SOURCE CONTRIBUTIONS

- Verified-zkEVM/ArkLib (Lean 4 formally verified cryptography library) — contributed multiple proofs merged into the library, including a STIR soundness lemma, general probability bounds (union bound, independence, i.i.d. powers), and lattice and finite-type utility lemmas.
- leanEthereum (leanVM, leanSig, leanSpec — Ethereum's post-quantum consensus client, built in Rust and Lean) — added regression tests covering field-canonicity and signature-verification security fixes, proposed a PQ VRF for leanEthereum after RANDAO breakdown, and corrected stale documentation.
- arkworks-rs (algebra, spongefish) — added Fiat-Shamir proof-of-work transcript support and serialization support for Box<T> to core Rust zero-knowledge proving libraries.
- Zcash (ZcashFoundation/frost-tools, orchard) — implemented Ironwood (NU6.3) external-signing support for FROST threshold signatures.
- FhenixProtocol (cofhesdk, cofhe-contracts) — fixed a kill-switch gating gap in the FHE TaskManager and added regression test coverage for the decrypt-permit path.
- succinctlabs/flock — hardened a zero-knowledge verifier against malformed-proof exploits identified in a third-party security audit.
- NethermindEth/latticefold — implemented the ℓ2-norm check from a 2026 lattice-folding scheme paper (LatticeFold+).
- noir-lang/awesome-noir — added noir-rlwe-gadgets to the Noir ecosystem's curated resource list.

PUBLICATIONS

Ashlar: An Arithmetization-Oriented Hash from an R1CS-Optimal Squaring Degree Engine *Preprint with full validation artifact (2026)*

Quantum Soundness of Grinding in Hash-Based SNARKs *Preprint; identifies a soundness placement flaw in the deployed STIR protocol (2026)*

Real-World Applications of Artificial Intelligence and Blockchain in Healthcare *SSRN Pre-print*

Independent Domination in Subcubic Graphs *Journal of Combinatorial Optimization, Vol. 43, pp. 28–41 (2022)*

Investigating Decentralising ZK-Rollups: A Path to Decentralized Scalability for Ethereum *Research Proposal (Unpublished) (2024)*

EDUCATION

Bachelor of Computer Science | Sharif University of Technology, Tehran, Iran

Master of Decision Science and Knowledge Engineering | Tehran University *(Discontinued)*

SELECTED CERTIFICATIONS

Via Coursera — from leading universities and institutions:

- Cryptography I — Stanford University
- Introduction to Applied Cryptography — University of Colorado, Colorado Springs
- Applied Cryptography — University of Colorado, Colorado Springs
- DeFi Specialization — Duke University
- Academic English Writing Specialization — University of California
- Entrepreneurial Finance: Strategy and Innovation — Duke University
- Game Theory — Stanford University

LANGUAGES

Persian (Native) **English** (Professional Proficiency, TOEFL: 101)